

Exercise 1

let d_1 to be the distance between the base and COM 1

$$P_{C1} = \begin{cases} d_1 c_1 \\ d_2 s_1 \end{cases} \quad v_{C1} = \begin{cases} -d_1 s_1 \dot{q}_1 \\ d_1 c_1 \dot{q}_1 \end{cases} \quad T_1 = \frac{1}{2} I_1 \dot{q}_1^2 + \frac{1}{2} m_1 d_1^2 \dot{q}_1^2 = \frac{1}{2} \dot{q}_1^2 (I_1 + m_1 d_1^2)$$

$$P_{C1} = \begin{cases} l_1 c_1 + q_2 \cos(q_1 - \frac{\pi}{2}) \\ l_1 s_1 + q_2 \sin(q_1 - \frac{\pi}{2}) \end{cases} = \begin{cases} l_1 c_1 + q_2 s_1 \\ l_1 s_1 - q_2 c_1 \end{cases} \quad v_{C1} = \begin{cases} -l_1 s_1 \dot{q}_1 + q_2 c_1 \dot{q}_1 + \dot{q}_2 s_1 \\ l_1 c_1 \dot{q}_1 + q_2 s_1 \dot{q}_1 - \dot{q}_2 c_1 \end{cases}$$

$$T_2 = \frac{1}{2} I_2 \dot{q}_1^2 + \frac{1}{2} m_2 (l_1^2 \dot{q}_1^2 + q_2^2 \dot{q}_1^2 + \dot{q}_2^2 - 2 l_1 \dot{q}_1 \dot{q}_2)$$

$$M_{11} = I_1 + m_1 d_1^2 + I_2 + l_1^2 m_2 + m_2 q_2^2 = e_1 + e_2 q_2^2$$

$$M_{12} = -m_2 l_1 = e_3$$

$$M_{22} = m_2 = e_2$$

$$M = \begin{pmatrix} e_1 + e_2 q_2^2 & e_3 \\ e_3 & e_2 \end{pmatrix} \quad \text{Now i compute } c_i = \dot{q}^T C_i \dot{q}$$

$$C_1 = \begin{pmatrix} 0 & e_2 q_2 \\ e_2 q_2 & 0 \end{pmatrix} \Rightarrow c_1 = 2 \dot{q}_1 \dot{q}_2 e_2 q_2, \quad c_2 = -e_2 q_2 \dot{q}_1^2$$

$$U = g_0 (m_1 d_2 s_1 + m_2 [l_1 s_1 - q_2 c_1])$$

$$\frac{\partial U}{\partial q_1} = g_0 (m_1 d_2 c_1 + m_2 [l_1 c_1 + q_2 s_1]) \quad \frac{\partial U}{\partial q_2} = -g_0 m_2 c_1 = -g_0 e_2 c_1$$

$$= g_0 (e_4 c_1 + e_2 q_2 s_1)$$

$$\begin{cases} (e_1 + e_2 q_2^2) \ddot{q}_1 + e_3 \ddot{q}_2 - g_0 e_2 c_1 + 2 \dot{q}_1 \dot{q}_2 e_2 q_2 = \tau_1 \\ e_3 \ddot{q}_1 + e_2 \ddot{q}_2 + g_0 (e_4 c_1 + e_2 q_2 s_1) - e_2 q_2 \dot{q}_1^2 = \tau_2 \end{cases}$$

Now i write it in $Y \dot{C} = \tau$

$$\begin{bmatrix} \ddot{q}_1 & \ddot{q}_1 q_2^2 - g_0 c_1 + 2 \dot{q}_1 \dot{q}_2 q_2 & \ddot{q}_2 & 0 \\ 0 & \ddot{q}_2 + g_0 q_2 s_1 - q_2 \dot{q}_1^2 & \ddot{q}_1 & g_0 c_1 \end{bmatrix} \dot{C} = \tau$$

the min. number of dyn. coefficients is 4, since Y have 4 linearly independent columns.

↳ Y

Exercise 2

The gen. momentum is

$$M \dot{q} = \begin{cases} e_1 + e_2 q_2^2 \dot{q}_1 + e_3 \dot{q}_2 \\ e_3 \dot{q}_1 + e_2 \dot{q}_2 \end{cases} \quad \text{and} \quad \dot{q}_0(t) = \begin{cases} \frac{b\pi}{T} \sin(\frac{\pi}{T} t) \\ 0 \end{cases} \quad \text{so}$$

$$M(q_0) \dot{q}_0 = \begin{cases} e_1 + e_2 k^2 \frac{b\pi}{T} \sin(\frac{\pi}{T} t) \\ e_3 \frac{b\pi}{T} \sin(\frac{\pi}{T} t) \end{cases} \quad \text{hence}$$

$$h_1 = \int_0^T e_1 + e_2 k^2 \frac{b\pi}{T} \sin(\frac{\pi}{T} t) dt = e_1 T + 2 e_2 k^2 b$$

$$h_2 = \int_0^T e_3 \frac{b\pi}{T} \sin(\frac{\pi}{T} t) dt = 2 b e_3$$

Exercise 3

$p = M\dot{q}$ so $p_2 = M_2\dot{q}$ with $M_2 = \text{First } n_2 \text{ rows of } M$.

hence we want $M_2\dot{q} = c$. The objective function is

$$p^T \dot{q} = (M\dot{q})^T \dot{q} = \dot{q}^T M \dot{q} \quad \text{since } M \text{ is symmetric.}$$

$\uparrow$ weighted norm

To solve $p_2 = M_2\dot{q}$ we need the pseudo inverse, and we use the weighted version with M , hence the sol. is:

$$\dot{q} = \bar{M}^{-1} M_2^T (M_2 \bar{M}^{-1} M_2^T)^{-1} c \quad \text{For the PR example, } n_2 = 1 \text{ so}$$

$$M_2 = (e_1 + e_2 q_2^2 \quad e_3) \Rightarrow \dot{q} = \left(\frac{c}{e_2 q_2^2 + e_1} \quad 0 \right)^T$$

Exercise 4

i use y to identify the task vector and y_1, y_2 its component. The prof in the exerc
ise use (x, y) as the component of the task vector, so take into account this difference

To impose the impedance model, we need to feedback linearize in the task space, let $M_\tau = \bar{J}^T M \bar{J}$ $n_\tau = \bar{J}^T (g + c + M \bar{J} \bar{J}' \dot{\xi})$, $\begin{cases} F = \bar{J}^T \tau \\ \tau = \bar{J}^T F \end{cases}$

$$M_\tau \ddot{\xi} + n_\tau = F$$

We feedback linearize: $F = M_\tau a + n_\tau \Rightarrow \ddot{\xi} = a \Rightarrow$

$$\begin{cases} \ddot{\xi}_1 = a_1 & \text{i choose } a_1 = M_d^{-1} (-D_d \dot{\xi}_1 - K_d (\xi_1 - \xi_{1E}) + F_d) \\ \ddot{\xi}_2 = a_2 & \text{i choose } a_2 = -K_d \dot{\xi}_2 - K_p (\xi_2 - \xi_{2E}) \end{cases} \Rightarrow$$

$$\begin{cases} M_d \ddot{\xi}_1 + D_d \dot{\xi}_1 + K_d (\xi_1 - \xi_{1E}) = F_d \\ \ddot{\xi}_2 + K_d \dot{\xi}_2 + K_p (\xi_2 - \xi_{2E}) = 0 \end{cases} \text{ hence the COMPLETE control law is}$$

$$\tau = \bar{J}^T \left(\bar{J}^T (g + c + M \bar{J} \bar{J}' \dot{\xi}) + (\bar{J}^T M \bar{J}) \begin{pmatrix} M_d^{-1} (-D_d \dot{\xi}_1 - K_d (\xi_1 - \xi_{1E}) + F_d) \\ -K_d \dot{\xi}_2 - K_p (\xi_2 - \xi_{2E}) \end{pmatrix} \right)$$

Now let's analyze the final position, starting from ξ_2 :

$$\ddot{\xi}_2 + K_d \dot{\xi}_2 + K_p (\xi_2 - \xi_{2E}) = 0 \Rightarrow s^2 \xi_2 + s K_d \xi_2 + K_p (\xi_2 - \xi_{2E}) = 0 \Rightarrow \xi_2 (s^2 + K_d s + K_p) = K_p \xi_{2E}$$

$$\Rightarrow \xi_2(s) = \frac{K_p \xi_{2E}}{s^2 + K_d s + K_p} \quad \lim_{t \rightarrow \infty} \xi_2(t) = \lim_{s \rightarrow 0} \xi_2(s) = \frac{K_p}{K_p} \xi_{2E} = \xi_{2E}$$

on the vertical axis, the desired point is reached.

Now consider ξ_1

$$M_d \ddot{\zeta}_1 + D_d \dot{\zeta}_1 + K_d (\zeta_1 - \zeta_{1E}) = F_d \Rightarrow M_d \zeta_1 s^2 + D_d \zeta_1 s + K_d \zeta_1 = K_d \zeta_{1E} + F_d \Rightarrow \zeta_1 (M_d s^2 + D_d s + K_d) = K_d \zeta_{1E} + F_d$$

$$\zeta_1 = \frac{K_d \zeta_{1E} + F_d}{M_d s^2 + D_d s + K_d} \Rightarrow \lim_{t \rightarrow \infty} \zeta_1 = \frac{K_d \zeta_{1E} + F_d}{K_d}$$

Exercise 5

In absence of gravity the model is $M\ddot{q} + c = \tau$ hence $\ddot{q} = \bar{M}^{-1}(\tau - c)$

$$\begin{cases} (e_1 + e_2 q_2^2) \ddot{q}_1 + e_3 \ddot{q}_2 - g_0 e_2 c_1 + 2 \dot{q}_1 \dot{q}_2 e_2 q_2 = \tau_1 \\ e_3 \ddot{q}_1 + e_2 \ddot{q}_2 + g_0 (e_4 c_1 + e_2 q_2 s_1) - e_2 q_2 \dot{q}_1^2 = \tau_2 \end{cases} \quad q_2 \text{ should be } \kappa$$

$$M = \begin{pmatrix} e_1 + e_2 q_2^2 & e_3 \\ e_3 & e_2 \end{pmatrix} = \begin{pmatrix} e_5 & e_3 \\ e_3 & e_2 \end{pmatrix} \text{ where } e_1 + e_2 \kappa^2 = e_5 \text{ hence}$$

$$\ddot{q} = \frac{1}{e_2 e_5 - e_3^2} \begin{pmatrix} e_2 & -e_3 \\ -e_3 & e_5 \end{pmatrix} \begin{pmatrix} \tau_1 \\ \tau_2 + e_2 \kappa \dot{q}_1^2 \end{pmatrix} \quad \text{since } \dot{q}_2 = 0 \Rightarrow c_1 = 0$$

we have to find the bound for $\ddot{q}_1$ over q_1

$$\ddot{q}_2 = \frac{1}{e_2 e_5 - e_3^2} (e_5 \tau_2 + e_2 e_5 \kappa \dot{q}_1^2 - e_3 \tau_1) = 0 \Rightarrow \tau_2 = \frac{1}{e_5} \left(\tau_1 e_3 - e_2 e_5 \kappa \dot{q}_1^2 \right) =$$

$$\frac{1}{e_5} \left(\tau_1 e_3 - e_2 e_5 \kappa \frac{\pi^2}{T^2} \sin^2 \left(\frac{\pi}{T} t \right) \right)$$

$$\ddot{q}_1 = \frac{\pi^2}{T^2} \cos \left(\frac{\pi}{T} t \right) = \frac{1}{e_2 e_5 - e_3^2} \left[e_2 \tau_1 - e_2 \kappa \frac{\pi^2}{T^2} \sin^2 \left(\frac{\pi}{T} t \right) - e_3 \frac{1}{e_5} \left(\tau_1 e_3 - e_2 e_5 \kappa \frac{\pi^2}{T^2} \sin^2 \left(\frac{\pi}{T} t \right) \right) \right]$$

the max is in $t=0$

$$\frac{\pi^2}{T^2} = \frac{1}{e_2 e_5 - e_3^2} \left[e_2 \tau_1 - e_3 \frac{1}{e_5} \tau_1 e_3 \right] = \frac{\tau_1 \left(e_2 - \frac{e_3^2}{e_5} \right)}{e_2 e_5 - e_3^2} \leq \frac{\tau_{1M} \left(e_2 - \frac{e_3^2}{e_5} \right)}{e_2 e_5 - e_3^2}$$

solve for T

$$\frac{\pi^2}{T^2} = \frac{\tau_{1M} \left(e_2 - \frac{e_3^2}{e_5} \right)}{e_2 e_5 - e_3^2} \Rightarrow \sqrt{\frac{e_2 e_5 - e_3^2}{\tau_{1M} \left(e_2 - \frac{e_3^2}{e_5} \right) \pi^2}} = T \quad \checkmark \text{ min Feasible time}$$